

Data Science Masters

Pandas Basic — Assignment Answers

Q1. Create a Pandas Series containing 4, 8, 15, 16, 23, and 42, then print it.

A Pandas Series is a one-dimensional labeled array. The following code creates the required Series and prints it.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

data = [4, 8, 15, 16, 23, 42]
series = pd.Series(data)

print(series)
```

Q2. Create a list containing 10 elements and apply pandas.Series() on it.

The following example creates a list of 10 elements and converts it into a Pandas Series.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

my_list = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

series = pd.Series(my_list)

print(series)
```

Q3. Create a Pandas DataFrame containing Name, Age, and Gender, then print it.

The DataFrame contains the three records given in the assignment: Alice, Bob, and Claire.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Claire"],
    "Age": [25, 30, 27],
    "Gender": ["Female", "Male", "Female"]
}

df = pd.DataFrame(data)

print(df)
```

Q4. What is a DataFrame in Pandas and how is it different from pandas.Series? Explain with an example.

A DataFrame is a two-dimensional, tabular data structure consisting of rows and columns. A Series is a one-dimensional labeled data structure.

Difference:

- Series: one-dimensional; similar to a single column.
- DataFrame: two-dimensional; can contain multiple columns, with each column being a Series.

Example:

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

# Series example
age = pd.Series([25, 30, 27])
print("Series:")
print(age)

# DataFrame example
df = pd.DataFrame({
    "Name": ["Alice", "Bob", "Claire"],
    "Age": [25, 30, 27]
})

print("\nDataFrame:")
print(df)
```

Q5. What are some common functions used to manipulate data in a Pandas DataFrame? Give an example.

Common DataFrame functions include head(), tail(), info(), describe(), drop(), rename(), sort_values(), fillna(), dropna(), groupby(), and query().

Example: sort_values() can be used to arrange records according to a column. For example, the following code sorts students by age.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["Alice", "Bob", "Claire"],
    "Age": [25, 30, 27],
    "Gender": ["Female", "Male", "Female"]
})

sorted_df = df.sort_values(by="Age")

print(sorted_df)
```

Q6. Which of the following is mutable in nature: Series, DataFrame, Panel?

Pandas Series and DataFrame are mutable, meaning their values can be changed after creation. The older Pandas Panel data structure was also mutable, although Panel was deprecated and removed from modern Pandas versions.

For modern Pandas, the important structures are Series and DataFrame, and both are mutable.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

s = pd.Series([1, 2, 3])
s[0] = 100

df = pd.DataFrame({"A": [1, 2, 3]})
df.loc[0, "A"] = 100

print("Modified Series:")
print(s)

print("\nModified DataFrame:")
print(df)
```

Q7. Create a DataFrame using multiple Series. Explain with an example.

A DataFrame can be created by combining multiple Pandas Series as columns. The Series should have compatible indexes so that their values are aligned correctly.

Here, three Series—Name, Age, and Gender—are combined into one DataFrame.

Google Colab / Jupyter Notebook Code:

```
import pandas as pd

name = pd.Series(["Alice", "Bob", "Claire"])
age = pd.Series([25, 30, 27])
gender = pd.Series(["Female", "Male", "Female"])

df = pd.DataFrame({
    "Name": name,
    "Age": age,
    "Gender": gender
})

print(df)
```

Note: The assignment asks for the work to be created in a Jupyter notebook, uploaded to GitHub, and shared through the dashboard with a public repository.